



# 魅力统计

## Fascinating Statistics

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### 第三章 随机变量

### Chapter 3 Random Variables

## Chapter 3 Random Variables

- 3.1 Random Variables and Distributions
  - 3.1.1 Common Discrete Distributions
  - 3.1.2 Common Continuous Distributions
- 3.2 Numerical Characteristics of Random Variables
- 3.3 Law of Large Numbers and Central Limit Theorem



## 3.1 Random Variables and Distributions

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- Basic concepts like random experiment, sample space, random event, classical/geometric model of probability have been introduced in Chapter 2.
- However, when dealing with complex problems, we need a tool that allows for a unified study of random events.
- The goal is to simplify problems into functional operations, enabling the use of mathematical tools such as calculus.
- A random variable is such a tool: **it connects the real world with the mathematical world.**
  - Simply put, a random variable converts each possible outcome of a random experiment into a number.
  - For example, suppose an organization wants to survey public opinion on a proposal, asking whether people support or oppose it, by randomly sampling 50 individuals;
  - According to the classical model of probability, there are  $2^{50}$  possible outcomes.
  - However, if we use  $X$  to represent the number of people who support the proposal, then  $X$  is a random variable whose possible values belong to  $\{0, 1, 2, \dots, 50\}$ .
  - Therefore, random variables helps us to describe the problem in a precise and concise manner.

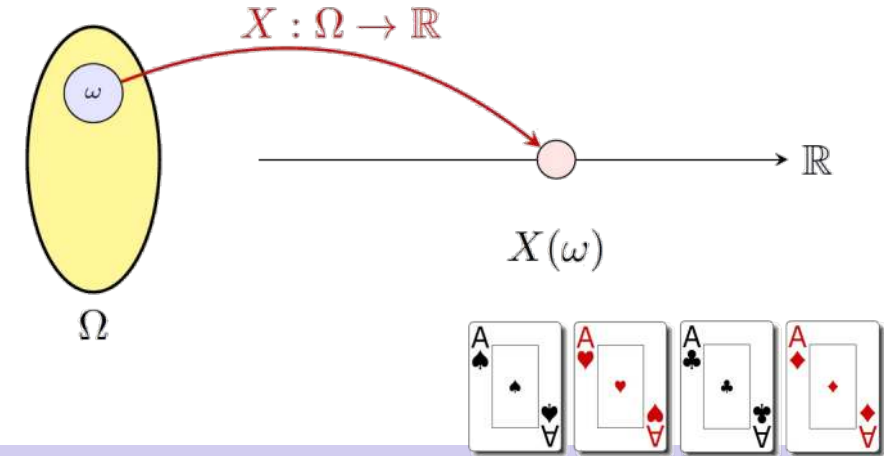


## 3.1 Random Variables and Distributions

- Here we provide the mathematical definition of a **random variable** (随机变量).

### Random Variable (simplified version)

A random variable (or r.v. for short) is a **real-valued function** defined on the sample space, typically denoted by capital letters like  $X, Y, Z$ .



### Example 3.1

- On Valentine's Day a restaurant offers a Lucky Lovers discount that could save couples money on their romantic dinners. How to define the Lucky Lovers discount as a random variable:
- When the waiter brings the check, he'll also bring the four aces from a deck of cards. He'll shuffle them and lay them out face down on the table. The couple will then get to turn one card over.
- If it's a black ace, they'll owe the full amount, but if it's the ace of hearts, the waiter will give them a \$20 Lucky Lovers discount.
- If they first turn over the ace of diamonds (hey—at least it's red!), they'll then get to turn over one of the remaining cards, earning a \$10 discount for finding the ace of hearts this time.



## 3.1 Random Variables and Distributions

### Solution

- It is not difficult to obtain the sample space of the game of Lucky Lovers:

$$\Omega = \{(\spadesuit A, \heartsuit A), (\heartsuit A, \clubsuit A), (\diamondsuit A, \heartsuit A), (\diamondsuit A, \spadesuit A), (\diamondsuit A, \clubsuit A)\}$$

- The Lucky Lovers discount can be defined as:

$$X(\omega) = \begin{cases} 20 & \text{if } \omega = \heartsuit A, \\ 10 & \text{if } \omega = (\diamondsuit A, \heartsuit A), \\ 0 & \text{otherwise.} \end{cases}$$

- It is a mapping from the outcomes in the sample space to numbers on the real line.

- With random variables defined, the study of random events can be simplified to the study of random variables.
- The study of r.v.s essentially involves examining all possible values that the r.v. can take and the probability associated with each value, which is known as the **probability distribution (概率分布)**.



## 3.1 Random Variables and Distributions

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- With the probability distribution, we can grasp the overall certainty of the random event, providing a foundation for further exploration of its underlying regularity.
- Based on the possible values a random variable can take, they can be classified into:
  - **Discrete random variable (离散型随机变量)**: take a finite or countable number of values, e.g., the number of customers that visit a restaurant.
  - **Continuous random variable (连续型随机变量)**: take continuous values (an uncountable number of values), e.g., the waiting time for a bus.
- The differences and similarities in how the probability distributions of these two types of r.v.s are described:
  - Discrete r.v. can be described using a **probability mass function (PMF, 概率质量函数)**.
  - Continuous r.v. can be described using a **probability density function (PDF, 概率密度函数)**.
  - Both types of r.v.s can be described using a **cumulative distribution function (CDF, 累积分布函数)**.



### 3.1.1 Common Discrete Distributions

#### Probability Mass Function

Let  $X$  be a discrete random variable,  $S = \{a_1, a_2, \dots\}$  be the **support** (支撑集) of  $X$ , i.e., the set of values that  $X$  can take. Then the **probability mass function** (PMF, 概率质量函数) of  $X$  is  $p: S \rightarrow [0, 1]$  defined as

$$p(a_i) = p_i = P(X = a_i), i = 1, 2, \dots,$$

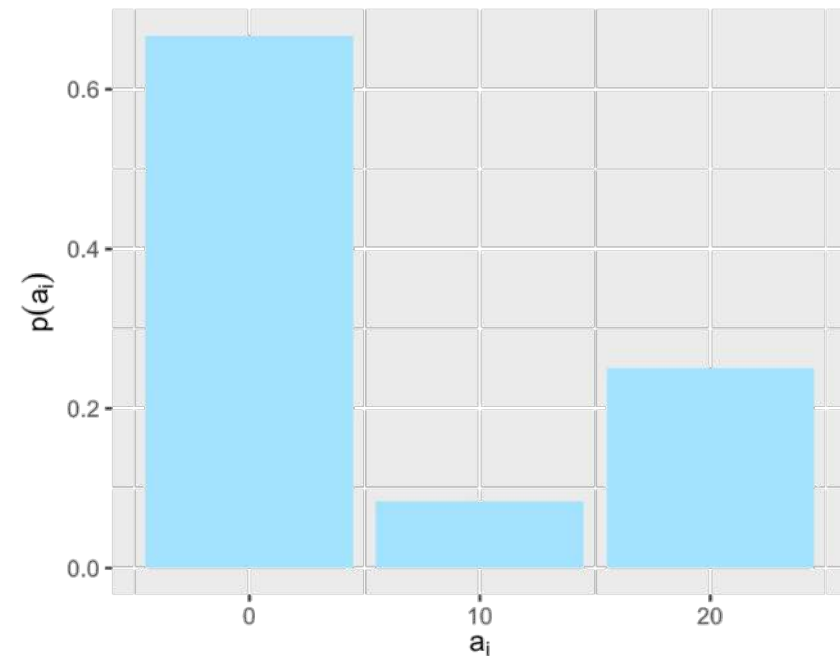
which satisfies:

- **Non-negativity** (非负性):  $p(a_i) \geq 0, i = 1, 2, \dots$ ;
- **Normalization** (规范性):  $\sum_i p(a_i) = 1$ .

- The PMF can also be displayed in a tabular format

|             |       |       |         |       |         |
|-------------|-------|-------|---------|-------|---------|
| Value       | $a_1$ | $a_2$ | $\dots$ | $a_i$ | $\dots$ |
| Probability | $p_1$ | $p_2$ | $\dots$ | $p_i$ | $\dots$ |

- The PMF can be displayed in a graphical format by plotting  $p(a_i)$  on the  $y$ -axis against  $a_i$  on the  $x$ -axis.



# 3.1.1 Common Discrete Distributions

## Example 3.1 (Continued)

Obtain the PMF of the Lucky Lovers discount.

## Solution

- It is not difficult to get:

$$P(\spadesuit A) = P(\heartsuit A) = P(\clubsuit A) = \frac{1}{4},$$

$$P(\{\diamond A, \heartsuit A\}) = P(\{\diamond A, \spadesuit A\}) = P(\{\diamond A, \clubsuit A\}) = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12}.$$

- Then:

$$P(X = 20) = P(\heartsuit A) = \frac{1}{4}, P(X = 10) = P(\{\diamond A, \heartsuit A\}) = \frac{1}{12}, P(X = 0) = \frac{1}{4} + \frac{1}{4} + \frac{1}{12} + \frac{1}{12} = \frac{2}{3}.$$

- Display the PMF in a tabular format:

|             |       |       |       |
|-------------|-------|-------|-------|
| Value       | 0     | 10    | 20    |
| Probability | 0.667 | 0.083 | 0.250 |



### 3.1.1 Common Discrete Distributions

- Bernoulli distribution (伯努利分布) is the simplest discrete distribution.

#### Bernoulli Trial and Bernoulli Distribution

- If a random experiment only have two possible outcomes,  $A$  and  $\bar{A}$ , then the experiment is called a **Bernoulli trial** (伯努利试验).
- If a random variable  $X$  only takes values 0 and 1, and  $P(X = 1) = p$ ,  $P(X = 0) = 1 - p$ , then  $X$  is called to follow a **Bernoulli distribution** (伯努利分布) with parameter  $p$ , denoted as
$$X \sim \text{Bernoulli}(p).$$

- The PMF of  $\text{Bernoulli}(p)$  can be expressed as
$$p(x) = p^x(1 - p)^{1-x}, x = 0, 1.$$

Jacob Bernoulli  
(1655-1705)



- The Bernoulli distribution is the foundation of many classical probability distributions, such as the binomial distribution (二项分布), the geometric distribution (几何分布), etc.



### 3.1.1 Common Discrete Distributions

#### $n$ -Fold Bernoulli Trial and Binomial Distribution

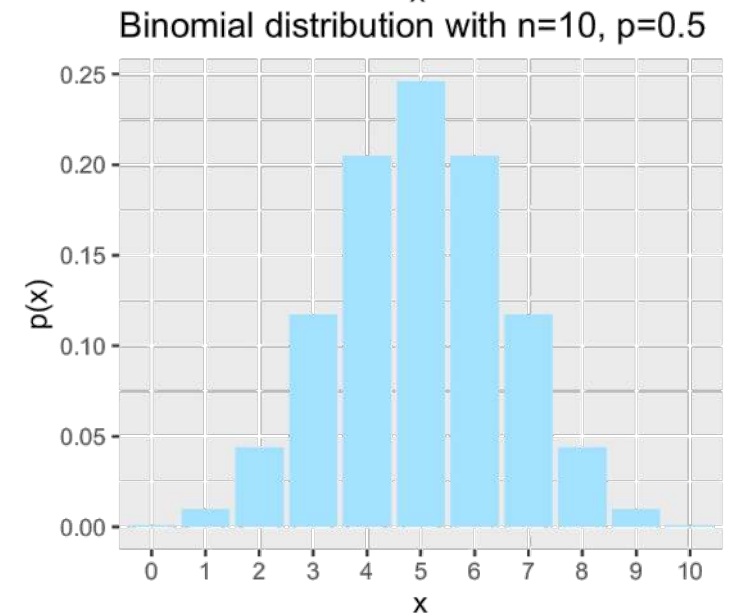
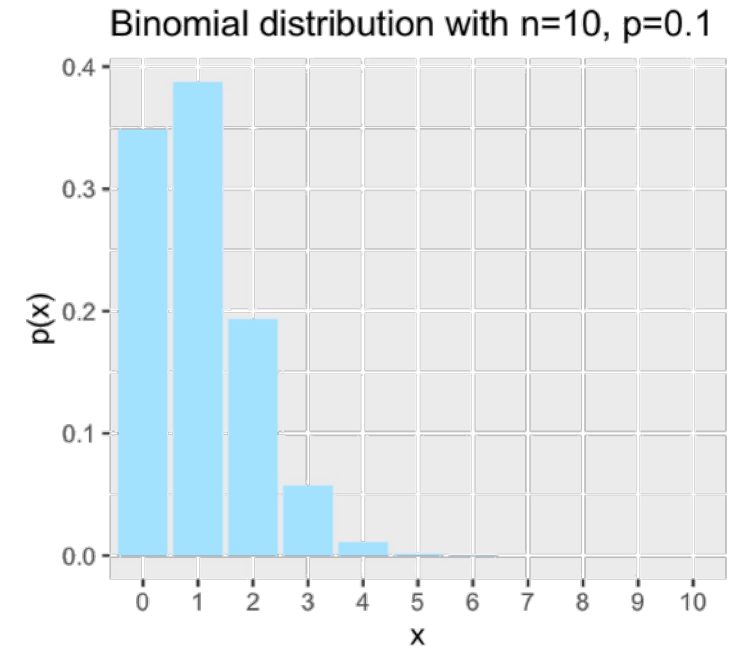
- An  $n$ -fold Bernoulli trial ( $n$ 重伯努利试验) is an experiment that repeats a Bernoulli trial  $n$  times independently. Note:
  - “independently” suggests that the result of each Bernoulli trial would not affect each other.
  - “repeat” suggests that the probability of event  $A$  in each Bernoulli trial, i.e.,  $P(A) = p$ , remains the same.

- Let  $X$  be a r.v. that records the number of times event  $A$  happens in an  $n$ -fold Bernoulli trial, then  $X$  is called to follow a **binomial distribution (二项分布)** with parameter  $n$  and  $p$ , denoted as

$$X \sim \text{Binomial}(n, p) \text{ (or } \text{Bin}(n, p), \text{B}(n, p)).$$

- The PMF of  $\text{Binomial}(n, p)$  can be derived as

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} = C_n^x p^x (1-p)^{n-x}, x = 0, 1, 2, \dots, n.$$



### 3.1.1 Common Discrete Distributions



#### Example 3.2

- A factory has 80 pieces of the same type of equipment, each operating independently, with a failure probability of 0.01.
- A single maintainer can only repair one piece of equipment at a time. The factory is considering two strategies for allocating maintainers:
  - Allocate 4 maintainers, with each responsible for maintaining 20 pieces of equipment.
  - Allocate 3 maintainer, with them jointly responsible for maintaining all 80 pieces of equipment.
- Please compare these two strategies in terms of the probability that a piece of equipment cannot be repaired in time when a failure occurs.



### 3.1.1 Common Discrete Distributions



#### Solution

- First consider the first strategy, let  $X_1, X_2, X_3, X_4$  be the number of equipment that fail at the same time among the 20 pieces maintained by the four maintainers, respectively. Then

$$X_i \sim \text{Binomial}(20, 0.01), i = 1, 2, 3, 4.$$

- Thus, the probability that a piece of equipment cannot be repaired in time is

$$\begin{aligned} P\left(\bigcup_{i=1}^4 \{X_i \geq 2\}\right) &\geq P\{X_1 \geq 2\} = 1 - P\{X_1 = 0\} - P\{X_1 = 1\} \\ &= 1 - \binom{20}{0} \times 0.01^0 \times 0.99^{20} - \binom{20}{1} \times 0.01^1 \times 0.99^{19} \approx 0.0169. \end{aligned}$$

- Then consider the second strategy, let  $X$  be the number of equipment that fail at the same time among the 80 pieces, then

$$X \sim \text{Binomial}(80, 0.01).$$

- Thus, the probability that a piece of equipment cannot be repaired in time is

$$P\{X \geq 4\} = 1 - \sum_{i=0}^3 P\{X = i\} = 1 - \sum_{i=0}^3 \binom{80}{i} \times 0.01^i \times 0.99^{80-i} \approx 0.0087. \quad \text{👍}$$

- Therefore, the second strategy is more efficient.



### 3.1.1 Common Discrete Distributions

#### Geometric Distribution

- Suppose that a Bernoulli trial is repeated independently until  $A$  occurs, let  $X$  be a r.v. that records the number of trials required, then  $X$  is called to follow a **geometric distribution (几何分布)** with parameter  $p$ , denoted as

$$X \sim \text{Geometric}(p) \text{ (or } \text{Geo}(p)\text{)}.$$

- The PMF of  $\text{Geometric}(p)$  can be derived as
$$p(x) = p(1 - p)^{x-1}, x = 1, 2, \dots$$
- The geometric distribution is the only discrete distribution which has the **memoryless property (无记忆性)**, i.e., for  $X \sim \text{Geo}(p)$  and any positive integers  $m, n$ , we have

$$P(X > m + n | X > m) = P(X > n).$$

#### Example 3.3

- A gambler at a casino is betting on 'big' in a game of 'big or small,' but the table has shown 'small' for ten consecutive rounds.
- The gambler believes that since 'small' has come up so many times already, the probability of another 'small' is very low.
- He is considering betting all his money on 'big' to try to win back his losses. What do you think?

Gambler's Fallacy  
(赌徒谬误)



### 3.1.1 Common Discrete Distributions

- Some probability distributions can be directly derived from the data generation mechanism itself, such as the Bernoulli distribution, binomial distribution, geometric distribution.
- Some other probability distributions are first derived mathematically or summarized from data, and later found to describe many real-world patterns.
- The Poisson distribution is an example of the second type which was first introduced by [Siméon-Denis Poisson](#) when he study the number of wrongful convictions in a given country.

#### Poisson Distribution

- Let  $X$  be a discrete r.v. with support  $\{0, 1, 2, \dots\}$ , if its PMF is

$$p(x) = P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, x = 0, 1, 2, \dots,$$

- where  $\lambda > 0$  is a constant, then  $X$  is said to follow a [Poisson distribution \(泊松分布\)](#) with parameter  $\lambda$ , denoted by  $X \sim \text{Poisson}(\lambda)$ .

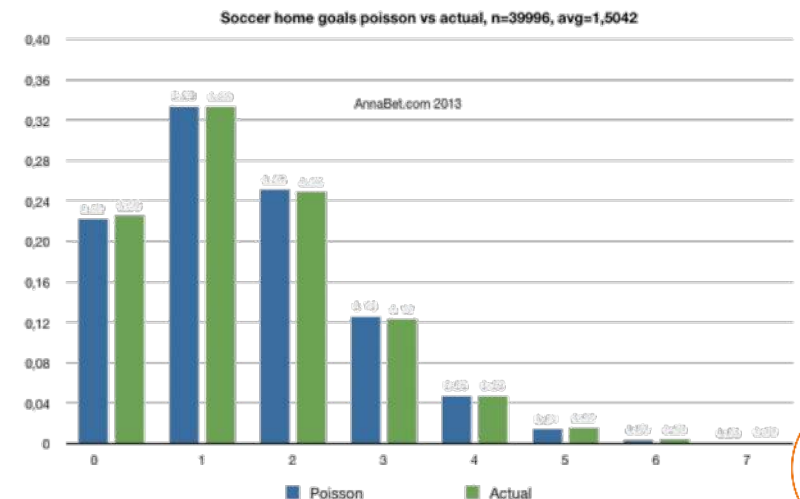


Siméon-Denis Poisson  
西莫恩·德尼·泊松  
(1781-1840)



## 3.1.1 Common Discrete Distributions

- While the Poisson distribution was initially introduced to solve a legal and criminal justice problem, it is widely used in various fields latter. Examples:
  - In 1860, an American astronomer fitted the Poisson distribution to the number of stars found in a space.
  - In 1898, a Russian economist showed that the frequency with which soldiers in the Prussian army accidentally killed by horse kicks could be well modeled by a Poisson distribution.
- Poisson distribution is used to describe the number of events occurring in a **fixed interval of time /space** if the events occur with a **constant rate** and **independently**. Examples:
  - The number of customers entering a store each day.
  - The number of traffic accidents at an intersection each year.
  - The number of typos on a page of a book.
  - The number of goals in a World Cup soccer match.
  - The number of mutations on a strand of DNA per unit length.



### 3.1.1 Common Discrete Distributions

#### Example 3.4

- During the World War II, were Germany's V2 missile capable of relatively precise target attacks?
- In World War I, Germany had already developed the "Paris Gun", which had a range of up to 113 kilometers.
- By World War II, especially in the later stages, German scientists had deployed the powerful V2 rocket in combat.
- Many V2 rockets were launched at various Allied nations, with 1,358 of them targeting London.
- Several analysts from British intelligence plotted all V2 rocket attacks on a map and found that the bombed areas showed a clear clustering trend.
- This indicated that the intelligence previously obtained about the performance of the V2 rockets might be significantly inaccurate, suggesting that the V2 rockets were more advanced than the intelligence had suggested.



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### 3.1.1 Common Discrete Distributions

Poisson distribution  
Flying-bomb hits on London



| number of hits, $k$               | 0      | 1      | 2     | 3     | 4    | 5 and over |
|-----------------------------------|--------|--------|-------|-------|------|------------|
| number of districts with $k$ hits | 229    | 211    | 93    | 35    | 7    | 1          |
| Poisson prediction                | 226.74 | 211.39 | 98.54 | 30.62 | 7.14 | 1.57       |

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#### Example 3.4

- The analysis report shocked the Allied high command, and the V2 rocket had become a nightmare looming over them.
- After the war, in 1946, American statistician R. D. Clarke selected a 144-square-kilometer area in southern London, divided it into 576 grids of 1/4 square kilometers each, and recorded the number of V2 rockets in each grid.
- Clarke found that the frequency of the counts closely matched the frequency predicted by the Poisson distribution. Therefore, he concluded that the analysis report was incorrect and that the V2 rocket strikes were random.
- Later, after obtaining the true data on the V2 rockets, it became clear that it is impossible for the V2 rockets to achieve relatively precise target attacks. Clarke's analysis was correct.



### 3.1.1 Common Discrete Distributions

- The Poisson distribution can be derived as a limiting case of the binomial distribution.

**Proof:** Consider the number of events within a unit time interval  $[0, 1]$  and divide it into  $n$  subintervals:

$$\left[0, \frac{1}{n}\right], \left[\frac{1}{n}, \frac{2}{n}\right], \dots, \left[\frac{i-1}{n}, \frac{i}{n}\right], \dots, \left[\frac{n-1}{n}, 1\right].$$

Several assumptions are made:

1. Let  $n$  be large so that each subinterval is very short, making it impossible for two or more events to occur within any subinterval (i.e., either no event occurs or exactly one event occurs).
2. The probability of an event occurring is proportional to the length of the subinterval, i.e.,  $\lambda/n$ .
3. Whether an event occurs in a subinterval is independent of the others.

Let  $X$  be the number of events within  $[0, 1]$ , by the assumptions above, we have  $X \sim \text{Binomial}\left(n, \frac{\lambda}{n}\right)$ . So

$$P(X = x) = \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}.$$

Let  $n \rightarrow \infty$ , it is not difficult to get

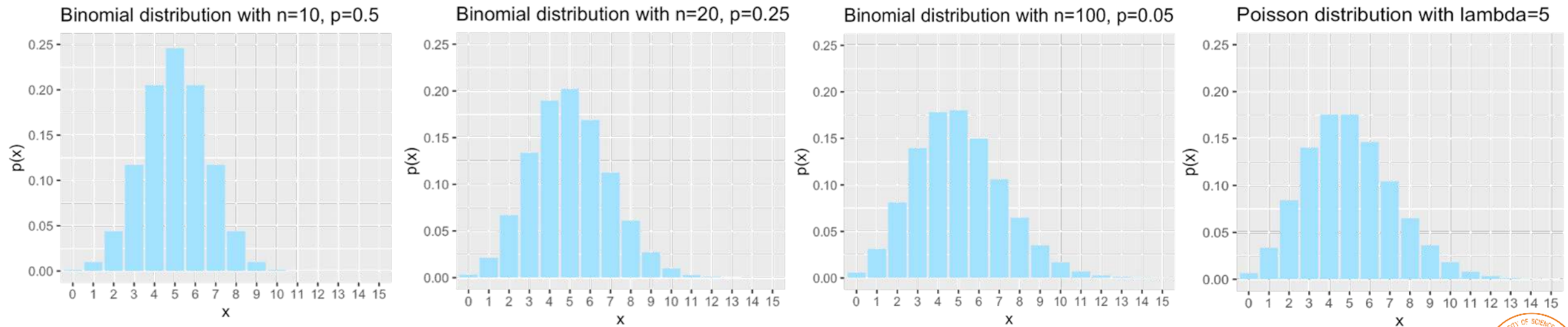
$$\lim_{n \rightarrow \infty} P(X = x) = \lim_{n \rightarrow \infty} \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} = \frac{\lambda^x}{x!} e^{-\lambda}.$$

Poisson Theorem  
(泊松定理)



### 3.1.1 Common Discrete Distributions

- The theorem suggests that for an  $n$ -fold Bernoulli trial with large  $n$  and small  $p$ , the binomial distribution  $\text{Binomial}(n, p)$  can be approximated by the Poisson distribution  $\text{Poisson}(np)$ .
- How large should  $n$  be and how small should  $p$  be?
- A rule of thumb is that when  $n > 100$  and  $p < 0.05$ , the Poisson distribution provides a good approximation to the binomial distribution.



### 3.1.1 Common Discrete Distributions



#### Example 3.5

- An insurance company has launched a life insurance policy where each participant is required to pay a premium of \$12 on January 1st.
- If a participant dies within the year, his/her family can receive a compensation of \$2,000 from the insurance company.
- Suppose 2,500 people participate in this insurance, and the probability of death for each person within the year is 0.002.
- What is the probability that the insurance company's profit from this life insurance policy is no less than \$20,000?



### 3.1.1 Common Discrete Distributions

#### Solution

- For the insurance company to make a profit of no less than \$20,000 from this life insurance policy, the maximum number of death is

$$\frac{2500 \times 12 - 20000}{2000} = 5.$$

- Let  $X$  be the number of death among the 2500 participants, then  $X \sim \text{Binomial}(2500, 0.002)$ . We would like to compute

$$P(X \leq 5) = \sum_{k=0}^5 \binom{2500}{k} 0.002^k 0.998^{2500-k}.$$

- Since  $n = 2500 > 100$  and  $p = 0.002 < 0.05$  in the binomial distribution, we apply the Poisson theorem, i.e., use  $\text{Poisson}(2500 * 0.002) = \text{Poisson}(5)$  to approximate  $\text{Binomial}(2500, 0.002)$ :

$$P(X \leq 5) \approx \sum_{k=0}^5 \frac{5^k}{k!} e^{-5} \approx 0.616.$$



## Chapter 3 Random Variables

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## 3.1.2 Common Continuous Distributions

### Continuous Random Variable and Probability Density Function

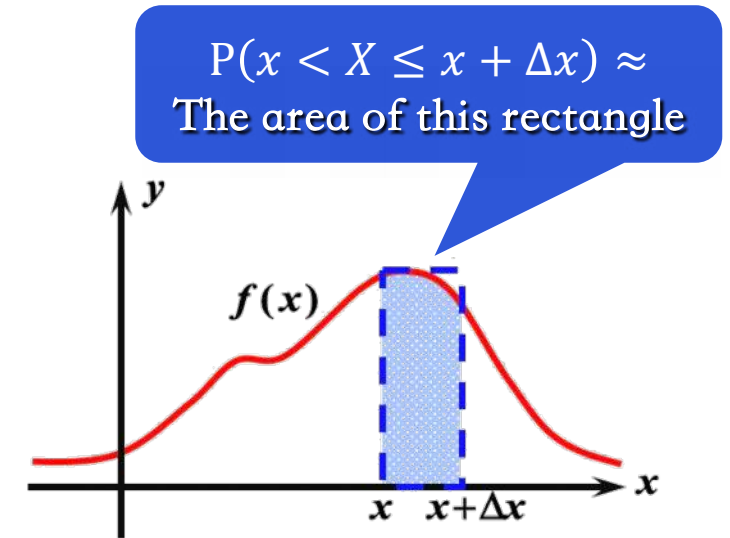
$X$  is called a continuous random variable if there exists a non-negative function  $f$ , defined for all  $x \in \mathbb{R}$ , satisfies that

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

for any  $-\infty < a \leq b < \infty$ . The function  $f$  is called the **probability density function (PDF, 概率密度函数)** of  $X$ .

It is not difficult to see that the PDF, like the PMF, has these properties:

- **Non-negativity (非负性):**  $f(x) \geq 0, \forall x \in \mathbb{R}$ ;
- **Normalization (规范性):**  $\int_{-\infty}^{\infty} f(x)dx = 1$ .
- Note that for any  $x \in \mathbb{R}$ , we have  $P(X = x) = \int_x^x f(u)du = 0$ .
- Thus, different from the PMF,  $f(x)$  does not reflect the probability of  $X$  taking the value  $x$ .



- The larger  $f(x)$  is, the greater the probability that  $X$  takes a value near  $x$ .
- $f(x)$  reflects the degree to which the probability is concentrated around  $x$ .



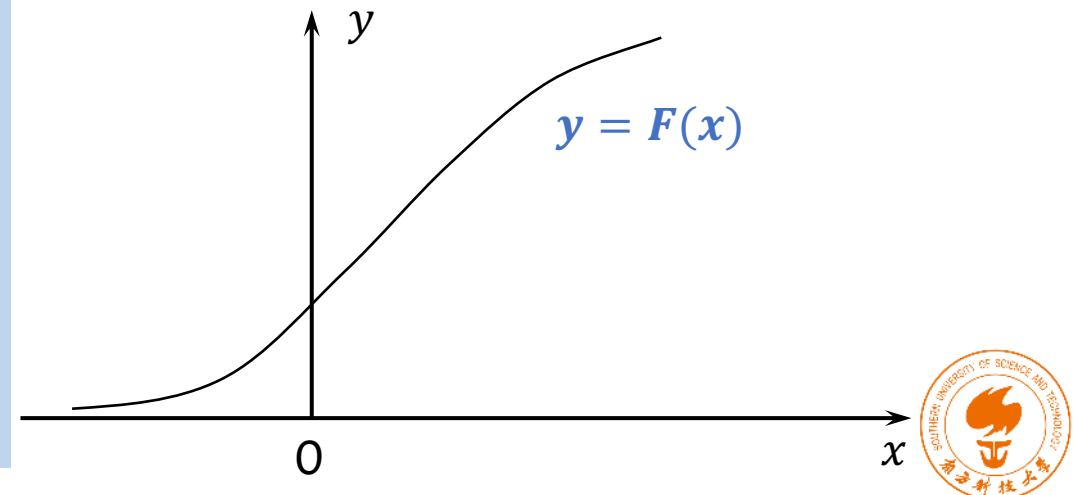
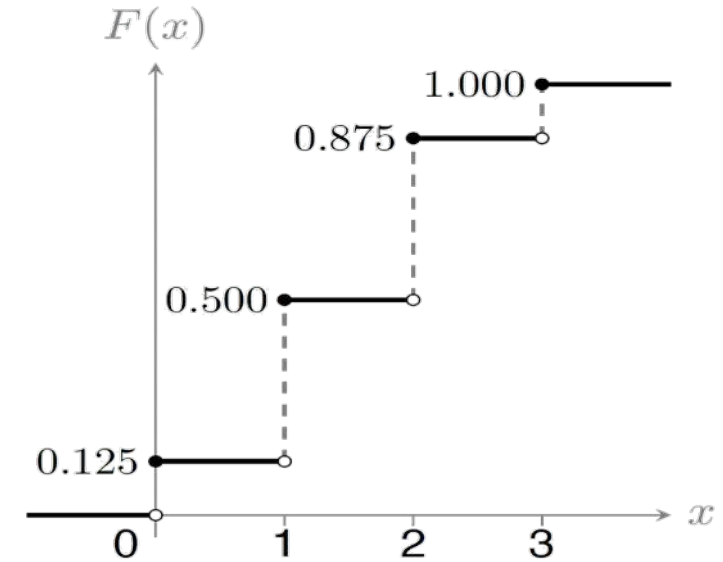
## 3.1.2 Common Continuous Distributions

### Cumulative Distribution Function

For a random variable  $X$ , either discrete or continuous, its **cumulative distribution function (CDF, 累积分布函数)** is defined as

$$F(x) = P(X \leq x), \forall x \in \mathbb{R}.$$

- For a **discrete r.v.**, its CDF is a step function (阶梯函数), which is  $F(x) = \sum_{a_i \leq x} p(a_i)$ .
- For a **continuous r.v.**, its CDF is a continuous function, which is  $F(x) = \int_{-\infty}^x f(u) du$ . Consequently,  $f(x) = F'(x)$ .
- The CDF is **non-decreasing** and **right-continuous**.
- The maximum of the CDF is  $F(\infty) = \lim_{x \rightarrow \infty} F(x) = 1$ .
- The minimum of the CDF is  $F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$ .
- For any real numbers  $a < b$ ,  $P(a < X \leq b) = F(b) - F(a)$ .



## 3.1.2 Common Continuous Distributions

- Uniform distribution (均匀分布) is the simplest continuous distribution.

### Uniform Distribution

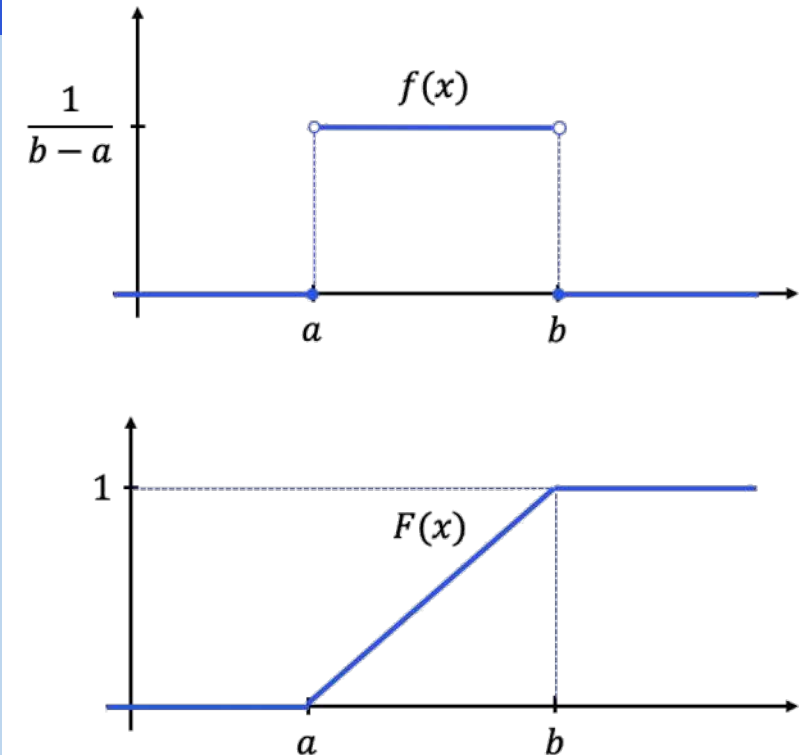
- If the probability density function (PDF) of a random variable  $X$  is

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b, \\ 0, & \text{otherwise} \end{cases}$$

- then  $X$  is said to follow a **uniform distribution (均匀分布)** on  $(a, b)$ , denoted as  $X \sim \text{Uniform}(a, b)$  or simply  $X \sim U(a, b)$ .

- The cumulative distribution function (CDF) of  $X \sim U(a, b)$  is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du = \begin{cases} 0, & \text{if } x \leq a \\ \frac{x-a}{b-a}, & \text{if } a < x < b. \\ 1, & \text{if } x \geq b \end{cases}$$



- For  $\forall (c, c + L) \in (a, b)$ ,  $P(c < X < c + L) = L/(b - a)$ , which only depends on the length but not the position of the interval, suggesting kind of “**equal likelihood (等可能性)**”.



## 3.1.2 Common Continuous Distributions

- The next common continuous distribution is the **exponential distribution** (指数分布).

### Exponential Distribution

- If the PDF of a random variable  $X$  is

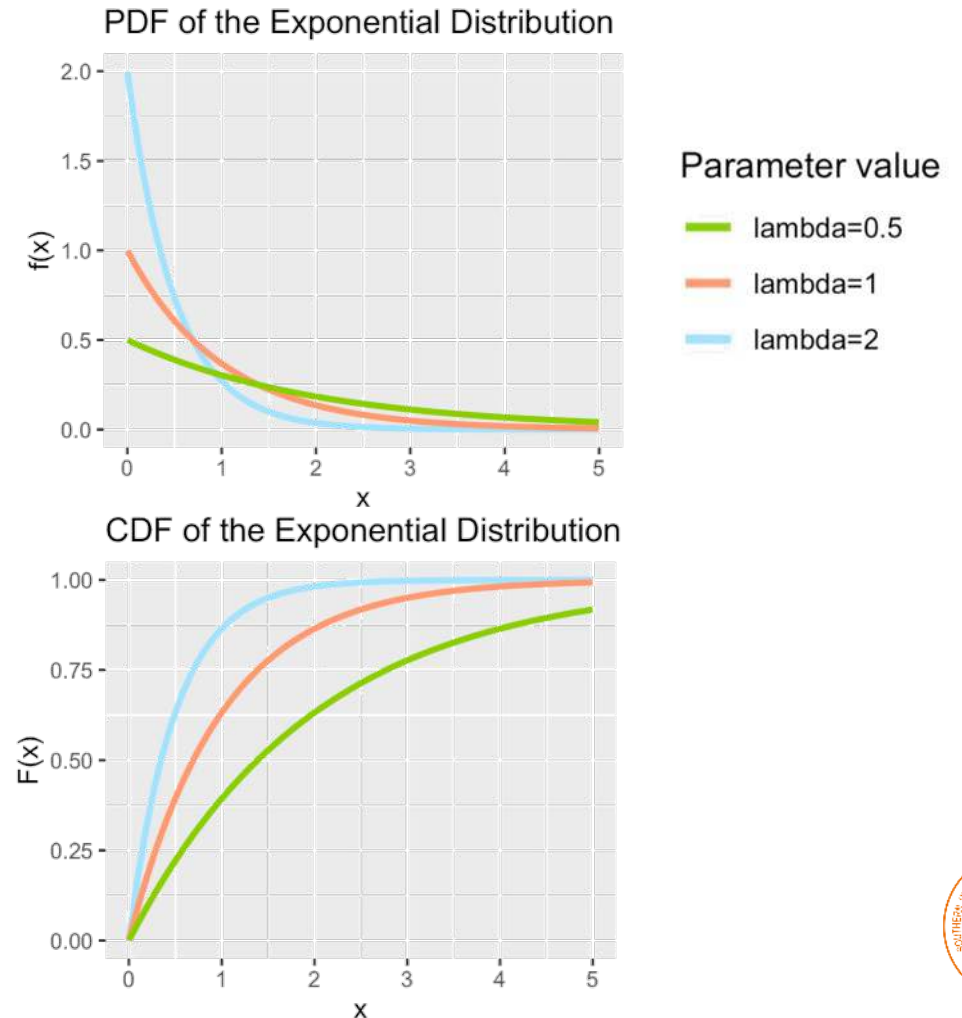
$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

- then  $X$  is said to follow an **exponential distribution** (指数分布) with parameter  $\lambda$ , denoted as  $X \sim \text{Exp}(\lambda)$ .

- The CDF of  $X \sim \text{Exp}(\lambda)$  is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du = \begin{cases} 1 - e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

- In practice, the exponential distribution often arises as the distribution of **the amount of time until some specific event occurs**.



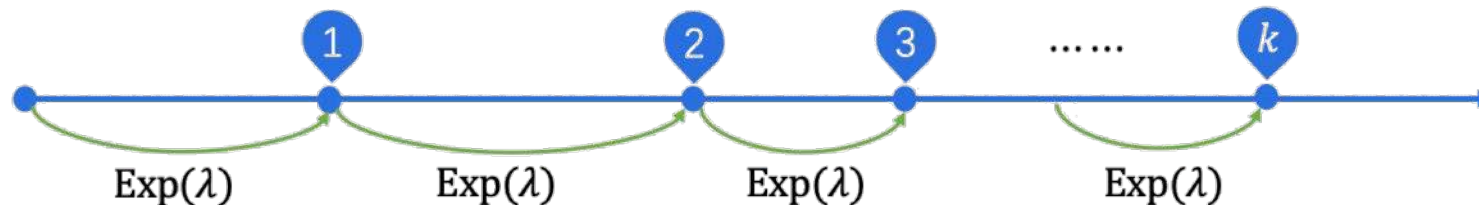
## 3.1.2 Common Continuous Distributions

- Have you noticed that the exponential distribution and the Poisson distribution use the same symbol  $\lambda$  for their parameter. Is there any relationship between them?
- In fact, the exponential distribution can be used to describe the distribution of the time intervals between events in a **Poisson process (泊松过程)**.

- A Poisson process can be simply understood as a process where random events occur **independently** and with a **constant rate** along the time axis.
- The number of events occurring within a unit time interval follows  $\text{Poisson}(\lambda)$ , then the number of events occurring in  $[0, t]$  follows  $\text{Poisson}(\lambda t)$ .
- Let  $X$  be the time until the first event occurs, then

$$P(X \leq t) = 1 - P(X > t) = 1 - P(\text{no event occurred in } [0, t]) = 1 - \frac{(\lambda t)^0}{0!} e^{-\lambda t} = 1 - e^{-\lambda t},$$

- which means that  $X \sim \text{Exp}(\lambda)$ .
- Similarly, we can show that the time intervals between events **independently** follow  $\text{Exp}(\lambda)$ .



## 3.1.2 Common Continuous Distributions

### Example 3.6

- There was an extraordinary rainstorm in Hong Kong on September 7th, 2023.
- The news reported that it was a once-in-500-years torrential rain.
- 500 year ago, it was the second year of the Jiajing reign of the Ming Dynasty (明朝嘉靖二年).
- Does it mean that this is the heaviest rain since the Ming Dynasty?



News: [https://www.hk01.com/article/939036?utm\\_source=01articlecopy&utm\\_medium=referral](https://www.hk01.com/article/939036?utm_source=01articlecopy&utm_medium=referral)



## 3.1.2 Common Continuous Distributions

### Answer

- In fact, the Observatory (天文台) describe the rain as a “once-in-100-years rain” and the Drainage Services Department (渠務署) referred to it as a “once-in-500-years rain”.
- The Observatory and the Drainage Services Department have different computation method.
- The Drainage Services Department’s computation is based on flood prevention standards.
- The Observatory’s “once-in-XXX-years” is the event rate of the Poisson process describing the event of rainfall with the same scale.
- Therefore, the time interval between two “once-in-100-years rain” follows  $\text{Exp}(1/100)$ .
- The expected time interval between two “once-in-100-years rain” is 100 years, but it doesn’t mean that we need to wait exactly 100 years to have one such rain.

其實新聞報道中常提及的「幾多年一遇」是表示概率。天文台與渠務署的計算公式亦不一樣。簡單而言，天文台的「多少年一遇」，是基於以往錄得的雨量及出現頻率作為統計基礎作出推算，是數據上的結論；而渠務署所指的「多少年一遇」，其實是防洪標準，一般市區排水幹渠系統足以應付重現期為200年一遇的暴雨，而今次「500年一遇」的暴雨侵襲，最後便導致多區出現水浸。



### 3.1.2 Common Continuous Distributions

- The exponential distribution is the only continuous distribution with the **memoryless property** (无记忆性), i.e., if  $X \sim \text{Exp}(\lambda)$ , then for any  $s, t > 0$ :

$$P(X > s + t | X > s) = P(X > t).$$

**Proof:** For any  $s, t > 0$ , since  $\{X > s + t\} \subset \{X > s\}$ , it follows that

$$P(\{X > s + t\} \cap \{X > s\}) = P(X > s + t) = 1 - F(s + t) = e^{-\lambda(s+t)}.$$

Then, by the definition of conditional probability:

$$P(X > s + t | X > s) = \frac{P(\{X > s + t\} \cap \{X > s\})}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

- The memoryless property greatly simplifies analysis, however, it makes the exponential distribution inappropriate for many real world applications.
- **Think:** Recall that the geometric distribution is the only discrete distribution with the memoryless property, is there any relationship between the geometric and exponential distributions?



## 3.1.2 Common Continuous Distributions

- The **normal distribution** (正态分布) is the most important distribution, without exception.

### Normal Distribution

- If the PDF of a random variable  $X$  is

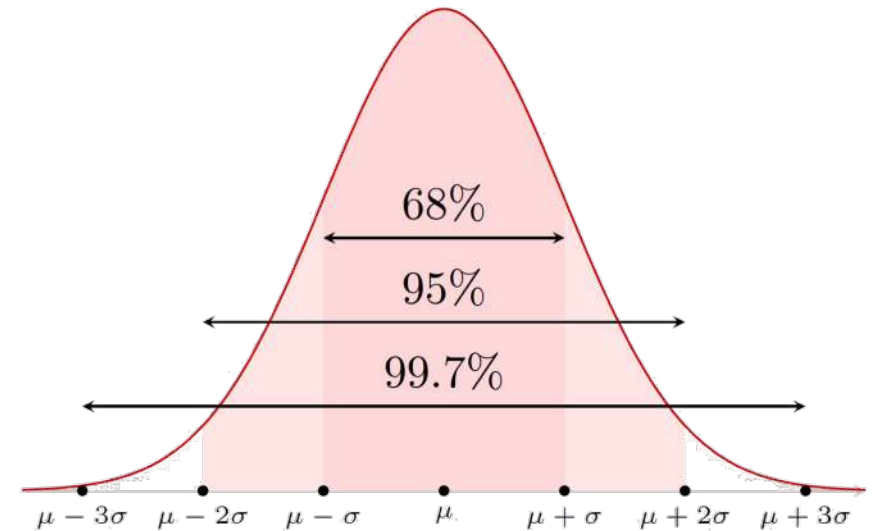
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty,$$

- then  $X$  is said to follow a **normal distribution** (正态分布) with parameter  $\mu$  and  $\sigma^2$  ( $\sigma > 0$ ), denoted as  $X \sim N(\mu, \sigma^2)$ .
- Specifically,  $N(0,1)$  is the **standard normal distribution** (标准正态分布), with PDF

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, -\infty < x < \infty.$$

- The CDF of  $X \sim N(0, 1)$  has no explicit expression, however, it is used very often and thus expressed as  $\Phi(x)$ :

$$\Phi(x) = P(X \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du.$$

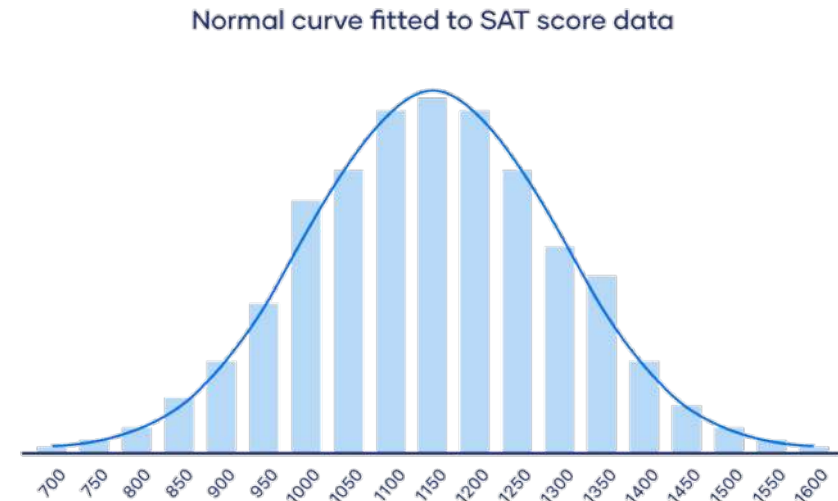
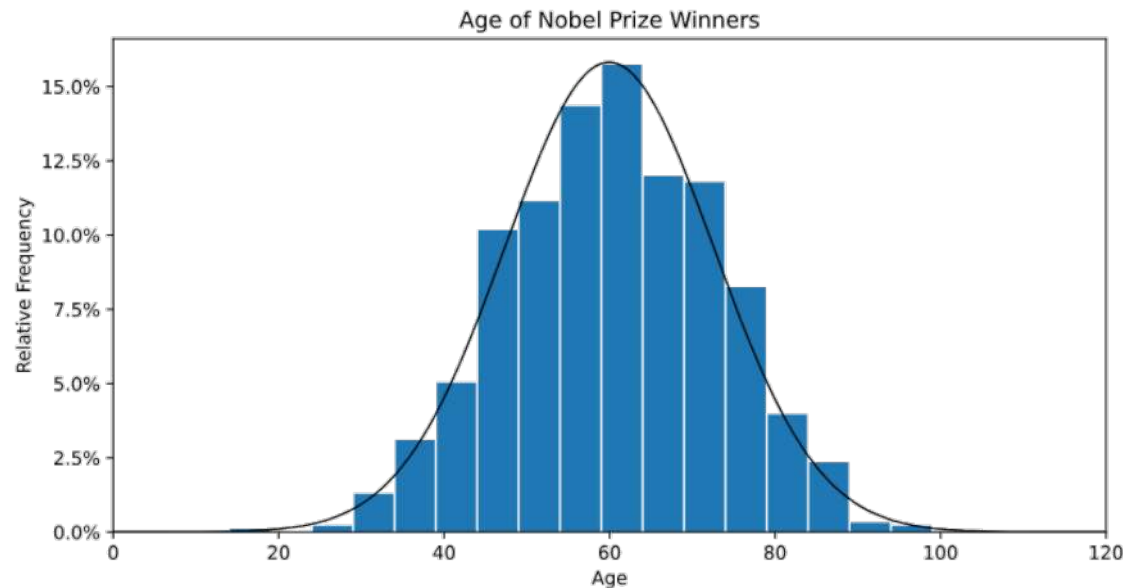


- The PDF of  $N(\mu, \sigma^2)$  is an elegant bell-shaped curve, symmetric about the parameter  $\mu$ .



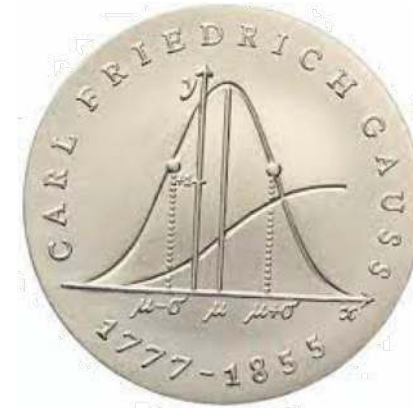
## 3.1.2 Common Continuous Distributions

- The normal distribution plays a vital role in Probability and Statistics, mostly because of the **Central Limit Theorem (CLT, 中心极限定理)**, which will be introduced in the next chapter.
- The CLT states that the sum/average of r.v.'s generally follows a normal distribution.
- Due to this fact, various fluctuations and measurement errors appear normally distributed.
- Moreover, normal distribution is often found to be a good model for weight, height, intelligence, temperature, pollution level, student grades, etc.



## 3.1.2 Common Continuous Distributions

- The normal distribution is also called the **Gaussian distribution** (高斯分布).

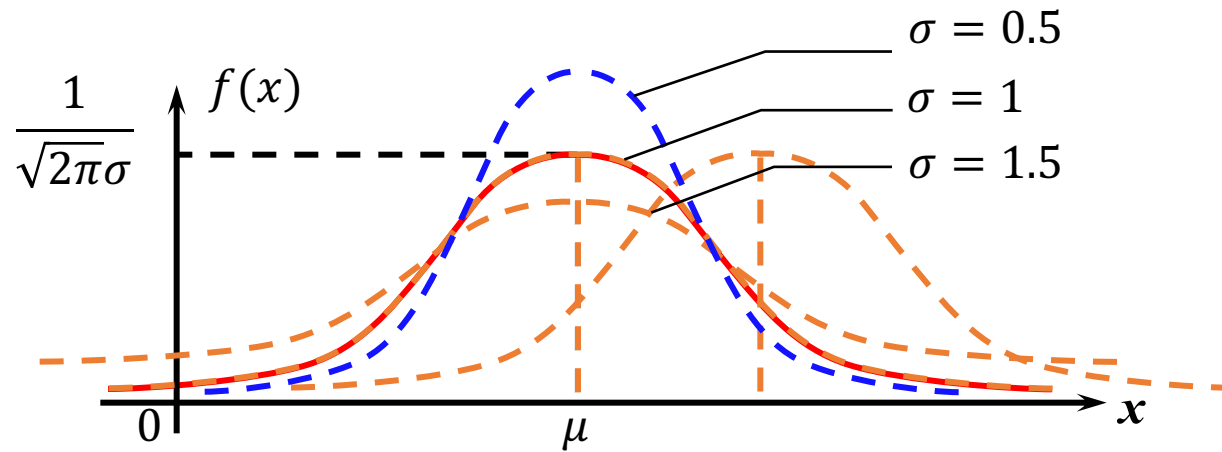


- Many people believe that it was Gauss who discovered the normal distribution.
- While Gauss did play a key role in establishing the significance of the normal distribution in history, he was not the first to propose the distribution.
- The French mathematician Poincaré suggested using the neutral term “normal distribution”.

正态分布的前世今生: <https://cosx.org/2013/01/story-of-normal-distribution-1/>

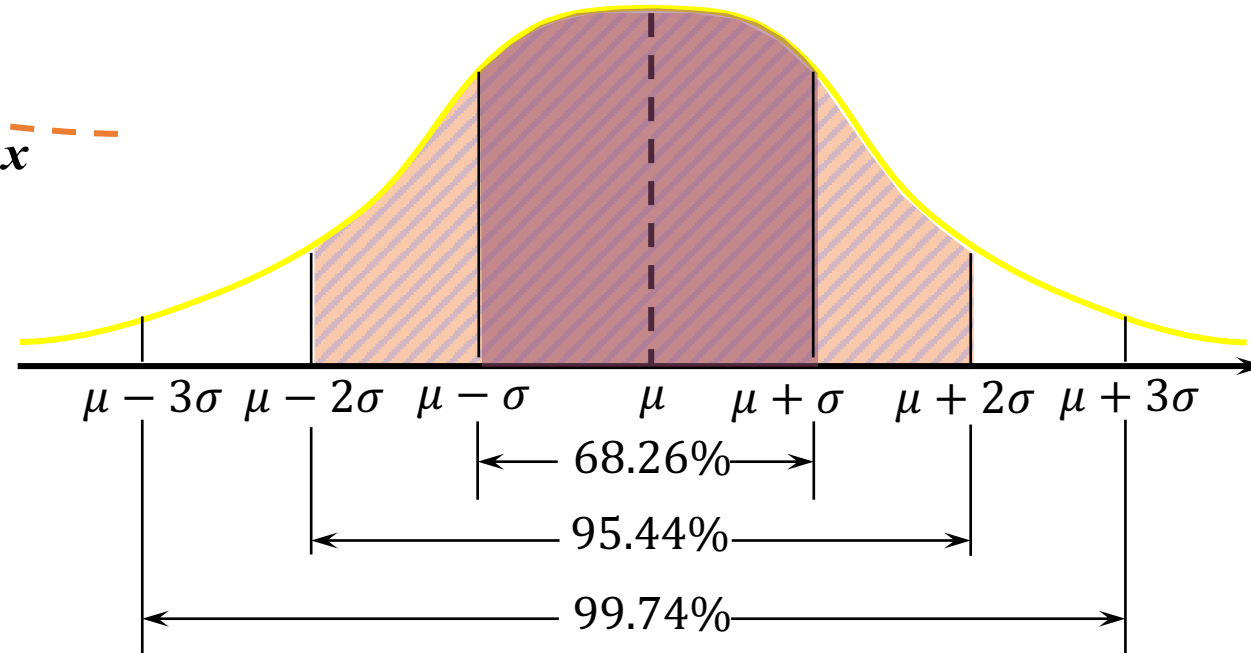


## 3.1.2 Common Continuous Distributions



- $\mu \uparrow$ : The PDF moves from the left to the right;
- $\mu \downarrow$ : The PDF moves from the right to the left;
- So,  $\mu$  determines the center of the distribution.
- $\sigma \uparrow$ : The PDF becomes flatter;
- $\sigma \downarrow$ : The PDF becomes sharper;
- So,  $\sigma$  determines the dispersion of the distribution.

- The three-sigma rule ( $3\sigma$ 原则): for a random variable  $X \sim N(\mu, \sigma^2)$ , the probability that its value falls into the interval  $(\mu - 3\sigma, \mu + 3\sigma)$  is almost 1.



- Also called the 68-95-99.7 rule.



## 3.1.2 Common Continuous Distributions

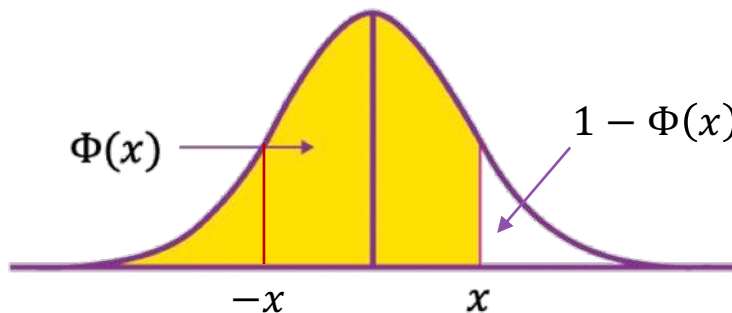
- If  $X \sim N(\mu, \sigma^2)$  and  $Z = (X - \mu)/\sigma$ , then  $Z \sim N(0,1)$  and consequently,

$$\begin{aligned} P(X \leq x) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) \\ &= P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right). \end{aligned}$$

Standardize (标准化)

- Moreover, by symmetry of the PDF of  $N(0,1)$ :

$$\Phi(-x) = 1 - \Phi(x), -\infty < x < \infty.$$



Standard Normal Probabilities

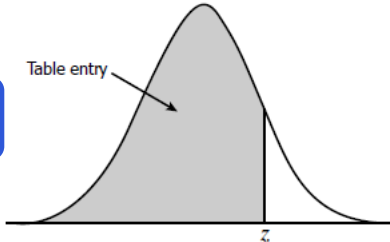


Table entry for  $z$  is the area under the standard normal curve to the left of  $z$ .

| $z$ | .00   | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| 0.1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| 0.2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| 0.3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| 0.4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| 0.5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| 0.6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| 0.7 | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| 0.8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| 0.9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |

- Since  $\Phi(x)$  does not have analytical expression, we would check its values in a probability table. E.g.,  $\Phi(1.96) = 0.975$ .



## 3.1.2 Common Continuous Distributions



### Example 3.7

- An expert witness (专家证人) in a paternity suit (亲子鉴定案) testifies that the length (in days) of human gestation (妊娠期) is approximately normally distributed with parameters  $\mu = 270$  and  $\sigma^2 = 100$ .
- The defendant in the suit is able to prove that he was out of the country during a period between 290 days and 240 days before the child was born.
- If the defendant is, in fact, the father of the child, what is the probability that the mother could have had the very long or very short gestation indicated in her testimony(证词)?

### Solution

- Let the mother's length of gestation be  $X$ , then  $X \sim N(270, 100)$ .
- The probability to be computed is  $P(\{X > 290\} \cup \{X < 240\})$ . Consider

$$P(X > 290) = P\left(\frac{X - 270}{10} > \frac{290 - 270}{10}\right) = P(Z > 2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228.$$

$$P(X < 240) = P\left(\frac{X - 270}{10} < \frac{240 - 270}{10}\right) = P(Z < -3) = 1 - \Phi(3) = 1 - 0.9987 = 0.0013.$$

- Therefore, the probability is  $0.0228 + 0.0013 = 0.0241$ .



## 3.1.2 Common Continuous Distributions



### Example 3.8

- A bus manufacturer is designing a bus. When determining the door height, they must ensure that it is not too high but also allows 99% of male passengers to pass through without bending.
- Assuming the height of all males (in cm) follows a normal distribution  $N(170, 36)$ , what should be the minimum door height to meet this requirement?

### Solution

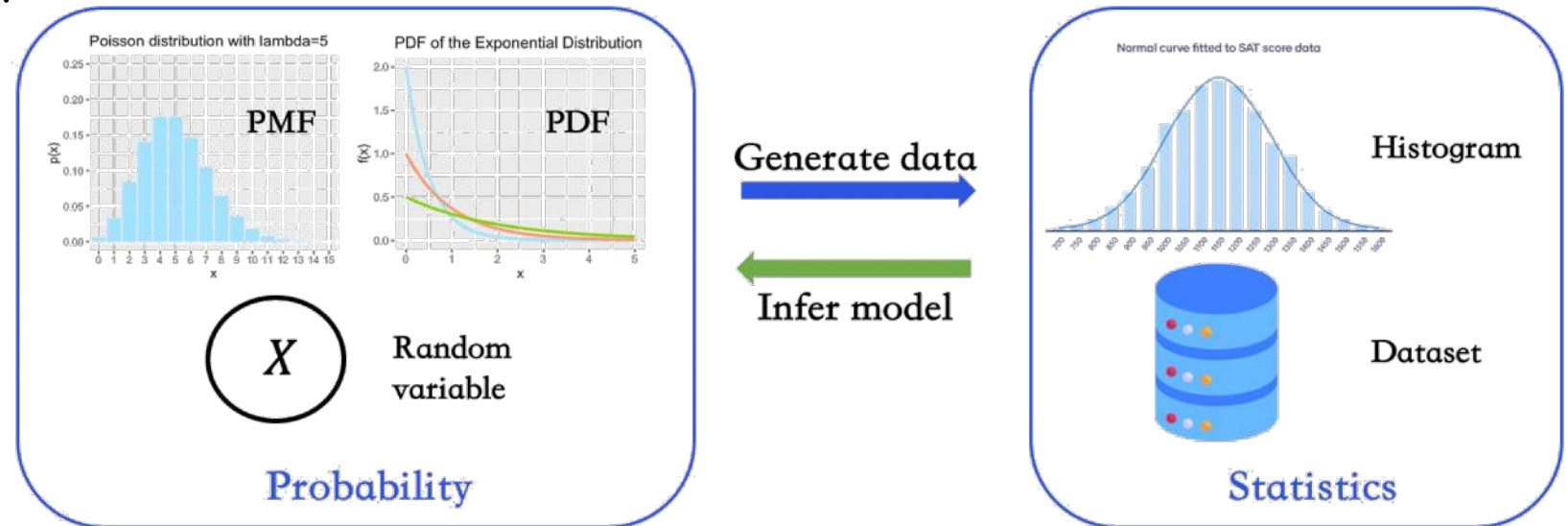
- Let  $X$  denote the height of a randomly selected male,  $h$  be the door height of the bus.
- Then the requirement can be expressed as  $P(X \leq h) \geq 0.99$ .
- Checking the probability table inversely, we find that for  $Z \sim N(0, 1)$ ,  $P(Z \leq 2.33) = 0.9901 > 0.99$ . So

$$\begin{aligned} P(X \leq h) &= P\left(\frac{X - 170}{6} \leq \frac{h - 170}{6}\right) = P\left(Z \leq \frac{h - 170}{6}\right) \geq 0.99. \\ \Rightarrow \frac{h - 170}{6} &\geq 2.33 \Rightarrow h \geq 170 + 13.98 \approx 184 \text{ cm.} \end{aligned}$$



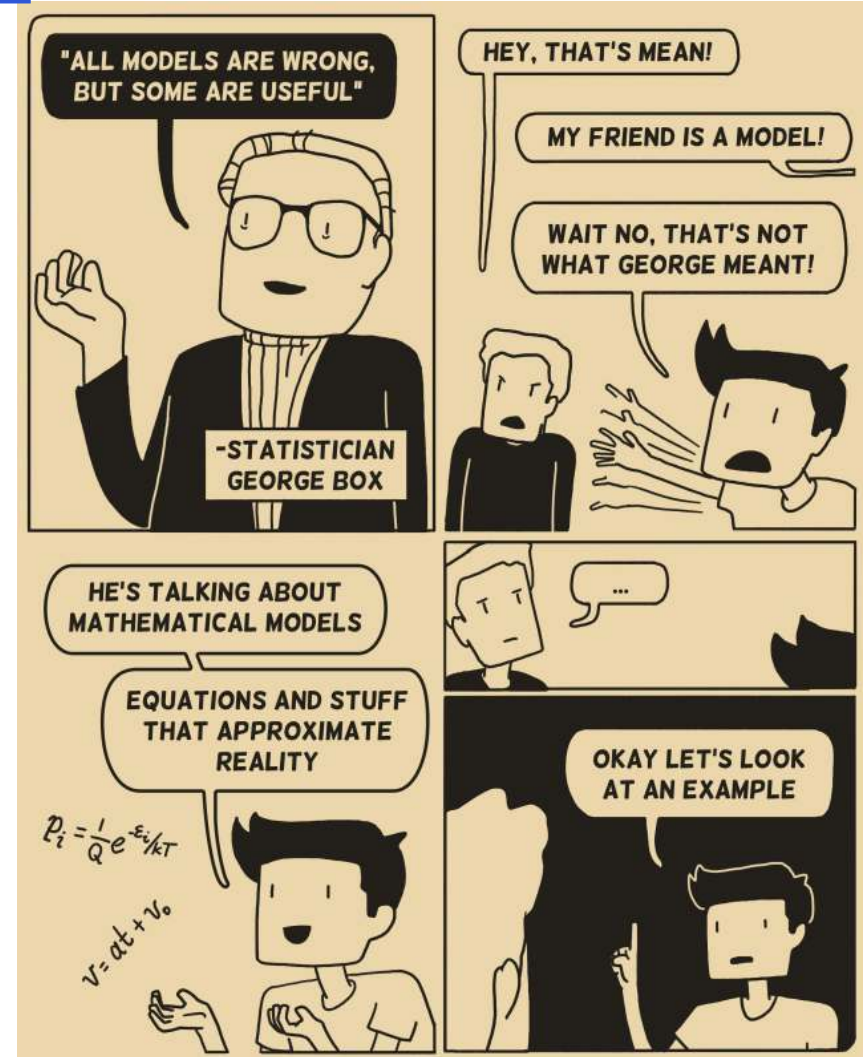
## 3.1 Random Variables and Distributions

- **Question:** What is the essence of a probability distribution?
- Typically, we can also plot a distribution based on the data we collect, e.g., using the **histogram** (直方图). How does this relate to the probability distribution mentioned?
  - The distribution plotted from collected data is just a form of data visualization.
  - A probability distribution does not correspond to a specific dataset, but is **a model**.
  - We can use this model to describe the mechanism of data generation or to summarize the patterns underlying the data.



## 3.1 Random Variables and Distributions

- Probability distributions are essentially models:
  - We usually assume a probability distribution model for the data, and then verify the assumption with the data collected.
  - It's hard to get a perfect match, but a certain range of deviation is acceptable.
  - “All models are wrong, but some are useful”.
  - In fact, the model itself is neither right nor wrong; the mistake lies in choosing the wrong model for a specific problem.
  - Probability distributions is like a toolbox, and each distribution is a specific tool inside the toolbox.
  - When faced with a problem, we need to find an appropriate tool from the toolbox to solve it.



概率分布之间的关系: <https://www.math.wm.edu/~leemis/chart/UDR/UDR.html>



## Chapter 3 Random Variables

- 3.1 Random Variables and Distributions
  - 3.1.1 Common Discrete Distributions
  - 3.1.2 Common Continuous Distributions
- 3.2 Numerical Characteristics of Random Variables
- 3.3 Law of Large Numbers and Central Limit Theorem



## 3.2 Numerical Characteristics of Random Variables

- A probability distribution provides a comprehensive and complete description of a r.v., however, sometimes we may want to describe the r.v. from specific aspects with just a few numbers.
- These numbers should be simple, clear, distinctive, and intuitive.
- The most commonly used numerical characteristics are the **mathematical expectation** (数学期望) and **variance** (方差).

### Mathematical Expectation

The **mathematical expectation** (数学期望) of a random variable  $X$  is denoted as  $E(X)$ :

- If  $X$  is a **discrete r.v.** with PMF  $p(x_k) = p_k = P(X = x_k), k = 1, 2, \dots$ , given  $\sum_{k=1}^{\infty} |x_k| p_k < \infty$ , then

$$E(X) \triangleq \sum_{k=1}^{\infty} x_k P(X = x_k) = \sum_{k=1}^{\infty} x_k p_k.$$

- If  $X$  is a **continuous r.v.** with PDF  $f(x)$ , given  $\int_{-\infty}^{\infty} |x| f(x) dx < \infty$ , then

$$E(X) \triangleq \int_{-\infty}^{\infty} x f(x) dx.$$

Also called the mean (均值), expectation (期望), expected value (期望值).



## 3.2 Numerical Characteristics of Random Variables

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- The concept of expectation can be traced back to Pascal and Fermat on the problem of points.
- They solved this problem based on a common principle: the value of future gains should be proportional to the chance of obtaining them.
- However, at that time, they did not formally propose a definition of expectation. The formal definition was introduced by Pierre Laplace in his book *Analytical Theory of Probability*.
- Simply put, the expectation is the **weighted average** of all possible values of a random variable.
- Expectation represents the **average result** or **long-term value** that we can anticipate from a series of random events, and it has wide applications in real life:
  - In the insurance industry, expectation can be used for setting the premium.
  - In the financial sector, investors use expectation to predict investment returns.



## 3.2 Numerical Characteristics of Random Variables



### Example 3.9

- In basketball, there are three ways to score – layups, mid-range shots, and three-pointers.
- Layups and mid-range shots have a higher shooting percentage but yield fewer points per shot; three-pointers have a lower shooting percentage but offer higher points per shot.
- Which scoring method is more efficient?

### Solution

- "More efficient" refers to long-term value, and this is where the concept of expectation becomes useful.
- The shooting percentage for layups is about 55%, so the expected points per attempt is
$$2 \times 55\% + 0 \times 45\% = 1.1.$$
- Similarly, the shooting percentage for mid-range shots is about 45%, making the expected points per attempt 0.9 points; the shooting percentage for three-pointers is about 35%, which gives an expected score of 1.05 points per attempt.
- Therefore, it is advisable to focus more on layups and three-point shots and less on mid-range shots.



## 3.2 Numerical Characteristics of Random Variables

### Solution

- This is not just a theoretical result. In the NBA, many teams are built around this strategy, emphasizing excellent centers or perimeter players who can drive to the basket, and stocking up on three-point shooters with defensive capabilities, focusing on efficient scoring from layups and three-point shots.
- Fans in China refer to this strategy as “魔球打法”, a typical example of which was the Houston Rockets during the James Harden era. Statistical data shows that in the 2018-2019 season, Harden took only 135 mid-range shots compared to 1,165 three-point attempts and 873 layups.
- A related concept, “Moneyball theory (魔球理论)”, is an operational system for sports teams, initially applied to baseball team management.
- Its core idea is to select undervalued players based on data analysis to achieve greater success with less investment.
- This theory has been popularized through a book and a movie, the latter titled *Moneyball* (点球成金 in Chinese), starring Brad Pitt.



## 3.2 Numerical Characteristics of Random Variables

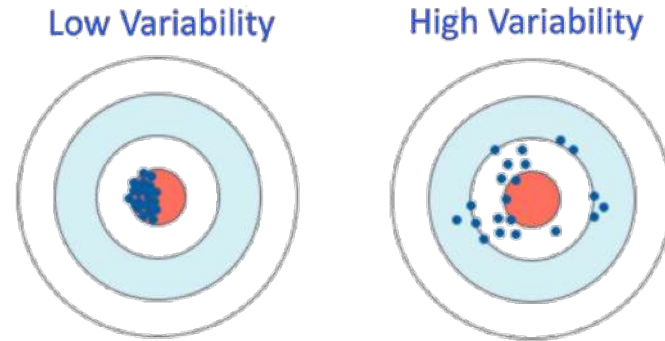
- While expectation represents the **average result**, we are also interested in the **stability/variability** (稳定性/波动性) of the result.
- **Variance (方差)** describes the variability of a random variable around its expectation.

### Variance and Standard Deviation

If a r.v.  $X$  satisfies that  $E(X^2) < \infty$ , then

$$\text{Var}(X) \triangleq E(X - E(X))^2$$

is called the **variance (方差)** of  $X$ , and  $\text{SD}(X) \triangleq \sqrt{\text{Var}(X)}$  is called the **standard deviation (SD, 标准差)**.



Q: Why not use  $E|X - E(X)|$  to describe the variability?

- Specifically, by the definition of expectation:
  - If  $X$  is a **discrete r.v.** with PMF  $p(x_k) = p_k = P(X = x_k)$ ,  $k = 1, 2, \dots$ , then  $\text{Var}(X) = \sum_{k=1}^{\infty} [x_k - E(X)]^2 \cdot p_k$ .
  - If  $X$  is a **continuous r.v.** with PDF  $f(x)$ , then  $\text{Var}(X) = \int_{-\infty}^{\infty} [x - E(X)]^2 f(x) dx$ .



## 3.2 Numerical Characteristics of Random Variables

### Example 3.9 (Continued)

Which of the three ways of scoring in basketball is more stable?

### Solution

- When discussing stability, variance/standard deviation becomes useful.
- The shooting percentage for layups is about 55%, with an expected score of 1.1, resulting in a variance of  $(2 - 1.1)^2 \times 55\% + (0 - 1.1)^2 \times 45\% = 0.99$ .
- Similarly, the variances for mid-range shots and three-pointers can be calculated as 0.99 and 2.0475.
- It is evident that although the expected score for three-pointers is higher than that for mid-range shots, the scoring fluctuation per three-point attempt is greater, indicating a higher risk.
- In the 2018 NBA playoffs, the Houston Rockets advanced to the Western Conference Finals but were narrowly defeated by the Golden State Warriors in a tough seven-game series.
- During this game, the Rockets missed 27 consecutive three-point attempts, with star player James Harden making only 2 out of 13 three-point attempts. In contrast, Warriors' Stephen Curry made 7 out of 15 three-point attempts, and Kevin Durant made 5 out of 11.



## 3.2 Numerical Characteristics of Random Variables

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- The expectation and variance of a random variable can be understood as **strength (实力)** and **luck (运气)** in real life.
- There are some basic properties of the expectation and variance:
  - For any constants  $a, b$ ,  $E(aX + b) = aE(X) + b$ .
  - For any constants  $a, b$ ,  $\text{Var}(aX + b) = a^2\text{Var}(X)$ , and thus  $\text{SD}(aX + b) = |a|\text{SD}(X)$ .
  - $\text{Var}(X) = E(X^2) - [E(X)]^2$ .
  - For any random variables  $X_1, X_2, \dots, X_n$ , we have  $E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$ .
  - If  $X_1, X_2, \dots, X_n$  are independent, then  $\text{Var}(X_1 + X_2 + \dots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)$ .
- These properties can be proved using the definitions of expectation and variance.
- The independence of random variables can be simply understood as the values of random variables will not affect each other, and the strict mathematical definition is omitted here.



## 3.2 Numerical Characteristics of Random Variables

- Let's summarize the expectation and variance of common probability distributions.

| Distribution         | PMF/PDF                                                                                                   | Expectation         | Variance              |
|----------------------|-----------------------------------------------------------------------------------------------------------|---------------------|-----------------------|
| Bernoulli( $p$ )     | $p(x) = p^x(1-p)^{1-x}, x = 0, 1$                                                                         | $p$                 | $p(1-p)$              |
| Binomial( $n, p$ )   | $p(x) = \binom{n}{x} p^x(1-p)^{n-x}, x = 0, 1, 2, \dots, n$                                               | $np$                | $np(1-p)$             |
| Geometric( $p$ )     | $p(x) = p(1-p)^{x-1}, x = 1, 2, \dots$                                                                    | $1/p$               | $(1-p)/p$             |
| Poisson( $\lambda$ ) | $p(x) = P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, x = 0, 1, 2, \dots$                                 | $\lambda$           | $\lambda$             |
| Uniform( $a, b$ )    | $f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b \\ 0, & \text{otherwise} \end{cases}$         | $\frac{a+b}{2}$     | $\frac{(b-a)^2}{12}$  |
| Exp( $\lambda$ )     | $f(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$ | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$ |
| N( $\mu, \sigma^2$ ) | $f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$               | $\mu$               | $\sigma^2$            |



## 3.2 Numerical Characteristics of Random Variables

- Finally, I would like to emphasize that in the age of AI, it is not enough to merely recognize common probability distributions.
- One must also understand which distribution to apply in different scenarios and the underlying considerations guiding that choice.

LUMIA Lab

2025-09-17

### FlowRL: Matching Reward Distributions for LLM Reasoning

**Abstract** | We propose FlowRL: matching the **full reward distribution** via flow balancing instead of maximizing rewards in large language model (LLM) reinforcement learning (RL). Recent advanced reasoning models adopt reward-maximizing methods (e.g., PPO and GRPO), which tend to over-optimize dominant reward signals while neglecting less frequent but valid reasoning paths, thus reducing diversity. In contrast, we transform scalar rewards into a **normalized target distribution** using a learnable partition function, and then minimize the reverse KL divergence between the policy and the target distribution. We implement this idea as a flow-balanced

- The target distribution is chosen as the **Boltzmann distribution**.
- It is a popular distribution in statistical mechanics (统计力学).
- This is because the Boltzmann distribution is the one that **maximizes the entropy** (can be understood as diversity) given a constant expectation.



## Chapter 3 Random Variables

- 3.1 Random Variables and Distributions
  - 3.1.1 Common Discrete Distributions
  - 3.1.2 Common Continuous Distributions
- 3.2 Numerical Characteristics of Random Variables
- 3.3 Law of Large Numbers and Central Limit Theorem



### 3.3 LLN and CLT

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- Law of large numbers (大数定律, LLN) and central limit theorem (中心极限定理, CLT) are the two most important theoretical results in probability theory.
- They are the theoretical foundation of many statistical methods.
  - LLN is a set of theorems, e.g., Bernoulli's LLN、Chebyshev's LLN、Khinchin's LLN, weak LLN, strong LLN, etc.
  - CLT is also a set of theorems, there are various versions with different assumptions, e.g., de Moivre-Laplace CLT, Lindeberg-Lévy CLT, etc.
  - We will not go into these theoretical details, but introduce the core ideas and real-world applications.
- Actually, the idea of LLN has been mentioned in previous classes:
  - When introducing expectation, it is said that expectation represents the average result or long-term value that we can anticipate from a series of random events.



### 3.3 LLN and CLT

- LLN essentially states that when we observe the values of a random variable repeatedly, the arithmetic average of these values will approach the expected value of the random variable:

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n} \xrightarrow{n \rightarrow \infty} \mu.$$

Simplified description

- In particular, when a random variable follows a Bernoulli distribution, it suggests that the observed frequency of an event will converge to the probability of the event.
- LLN means that probability can be approximated by frequency and expectation can be approximated by arithmetic average.
- This is one of the basic ideas of parameter estimation methods in statistics.
- LLN has an important practical application — the Monte Carlo method (蒙特卡洛方法).
- The Monte Carlo method, also known as the statistical simulation method, is a numerical computation method that uses random sampling to estimate unknown quantities.



### 3.3 LLN and CLT

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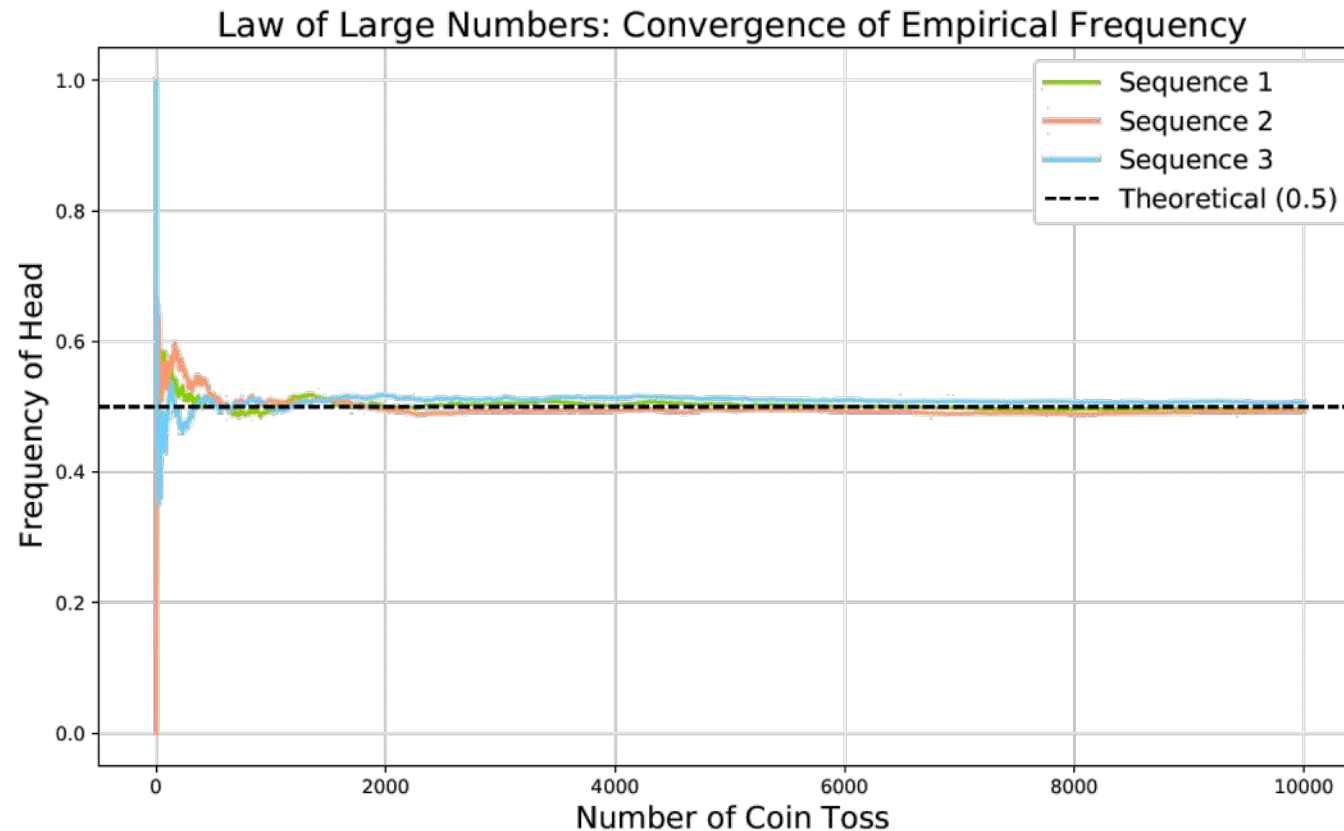
- In the 1940s, during World War II, the United States initiated the "Manhattan Project" to develop the atomic bomb.
- This marked the first time in human history that nuclear fission was manipulated to create an atomic bomb, a process with high risks, particularly due to the complex and highly random behavior of neutrons, which made precise calculations extremely challenging.
- The father of modern computers, the brilliant mathematician John von Neumann, along with others, proposed a random simulation method and named it after the famous casino in Monaco, Monte Carlo, which is called the Monte Carlo method.
- This technique is commonly used for complex integrations and probability calculations and finds extensive applications in fields such as macroeconomics, biomedical research, financial engineering, computational physics, and machine learning.



## 3.3 LLN and CLT

### Example 3.10

Use Python to simulate the results of a coin tossing experiment and see how the frequency of heads changes as the number of coin tosses increases?



## 3.3 LLN and CLT

### Example 3.11

- On June 12<sup>th</sup>, 2009, 5,141 families with financial difficulty in Wuhan participated in a public lottery for an affordable residential complex (经济适用住宅小区).
- As a result, 6 of the 124 lucky families had consecutive qualification certificate numbers (资格证明的编号是连续的).
- After investigation, the application materials of the 6 families were fake and their qualification was cancelled.



本页位置: 首页 → 新闻中心 → 国内新闻

### 武汉经适房6连号事件黑幕:委托人"好处费"近百万

2009年07月06日 10:25 来源:《瞭望》新闻周刊 [发表评论](#) 【字体: ↑大 ↓小】

“六连号”申购造假事件的背后,折射出经济适用住房政策的尴尬

文/《瞭望》新闻周刊记者

倍受关注的武汉市经济适用房“六连号”事件在十余天之后水落石出。

6月26日,由武汉市委、市政府成立,由纪检监察机关、司法机关等组成的工作组通报,这是一起由社会中介人员与有关部门工作人员相互勾结,利用经济适用房摇号进行舞弊、涉嫌经济犯罪的案件,王频等5名涉案人员已抓获到案,武汉市国土房产局副局长朱志强等5名国家工作人员因渎职、失职受到严肃处理。



### 3.3 LLN and CLT

#### Example 3.11



- Coincidentally, on July 29<sup>th</sup>, 2009, the lottery results of an affordable residential complex in Laohekou (老河口市, a city in Hubei) were posted online.
- People found that out of 1,138 eligible applicants, 514 were selected, in which 14 families had consecutive qualification certificate numbers.
- After multiple investigations, no illegal operations were found in this lottery.
- These two incidents inspired people's enthusiasm for calculating probabilities.
- What's the probabilities of obtaining 6 or more consecutive numbers when randomly selecting 124 out of 5,141 numbers? How about the case of obtaining 14 or more consecutive numbers when randomly selecting 514 out of 1,138 numbers?



### 3.3 LLN and CLT

#### Solution

- The problem can be described as: randomly select  $m$  numbers from  $N$  consecutive numbers with equal probability, find the probability of at least  $k$  consecutive numbers among the  $m$  numbers selected, denoted by  $P(N, m, k)$ .
- It seems to be a classical probability problem which can be solved by permutations and combinations.
- However, the calculation is very complicated.
- By adopting a different approach, we can try to implement the Monte Carlo method using Python to compute the probabilities.
- By the Monte Carlo method using Python, we obtain:

$$P(5141, 124, 6) \approx 0.00000,$$
$$P(1138, 514, 14) \approx 0.00824.$$

The results are different if you use another random seed

- Therefore, the probability of the incident in Wuhan is much lower than that in Laohekou.

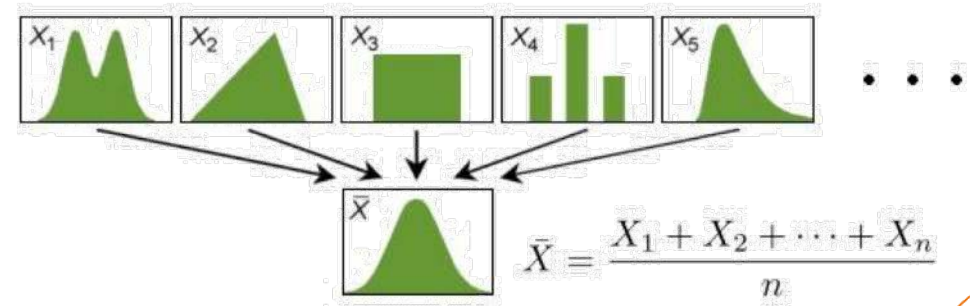


### 3.3 LLN and CLT

- CLT is also a theorem about the arithmetic average of a set of random variables:
  - CLT essentially states that **the arithmetic average of a set of random variables that meet certain requirements asymptotically follows a normal distribution:**

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{n \rightarrow \infty} N(0, 1) \text{ or } \bar{X} \xrightarrow{n \rightarrow \infty} N\left(\mu, \frac{\sigma^2}{n}\right).$$

- The CLT does not require  $X_1, X_2, \dots, X_n$  to follow any specific distribution, so it is a universal behavior across different probability distributions (with finite expectation and variance).
- All the complexity and chaos are dissolved under the mysterious curve of the normal distribution.
- Initially, mathematicians refer to this theorem as the Limit Theorem. However, due to its importance in probability theory, the word “central” was added.

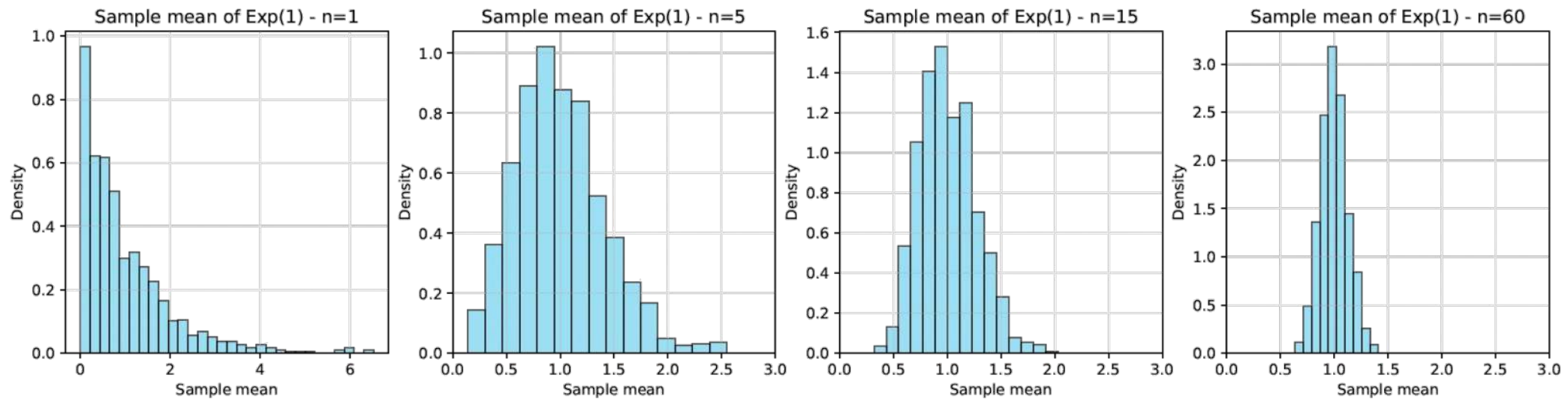


### 3.3 LLN and CLT

- The CLT explains why many measures in reality are normally distributed: they are typically the combined effect of multiple factors.

#### Example 3.12

Use Python to simulate and see how the arithmetic mean of multiple random variables that follow an exponential distribution converges to a normal distribution?



- Rule of thumb: when  $n \geq 30$ ,  $\bar{X}$  can be well approximated by a normal distribution, however, it depends on the specific distribution of  $X_1, X_2, \dots, X_n$ .



# Probability Theory and Statistics

- The first three chapters are about Probability Theory, in the following chapters, we will enter the field of Statistics.
- What's the relationship between Probability Theory and Statistics?
- Simply put, **probability theory and statistics study opposite problems**:
  - In Probability Theory, we are given the probability model and parameters of a random experiment, the question of interest is typically calculating the probability of a certain outcome.
  - In Statistics, the probability model or parameters of a random experiment are usually unknown, and we have collected some data from the experiment based on which the probability model and its parameters can be inferred, helping us understand the mechanism that generates the data.



Probability Theory



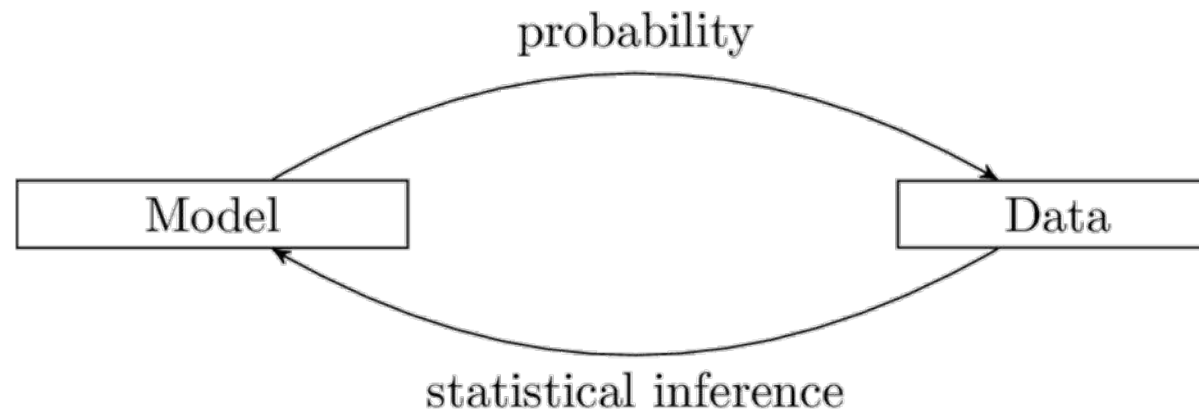
Statistics

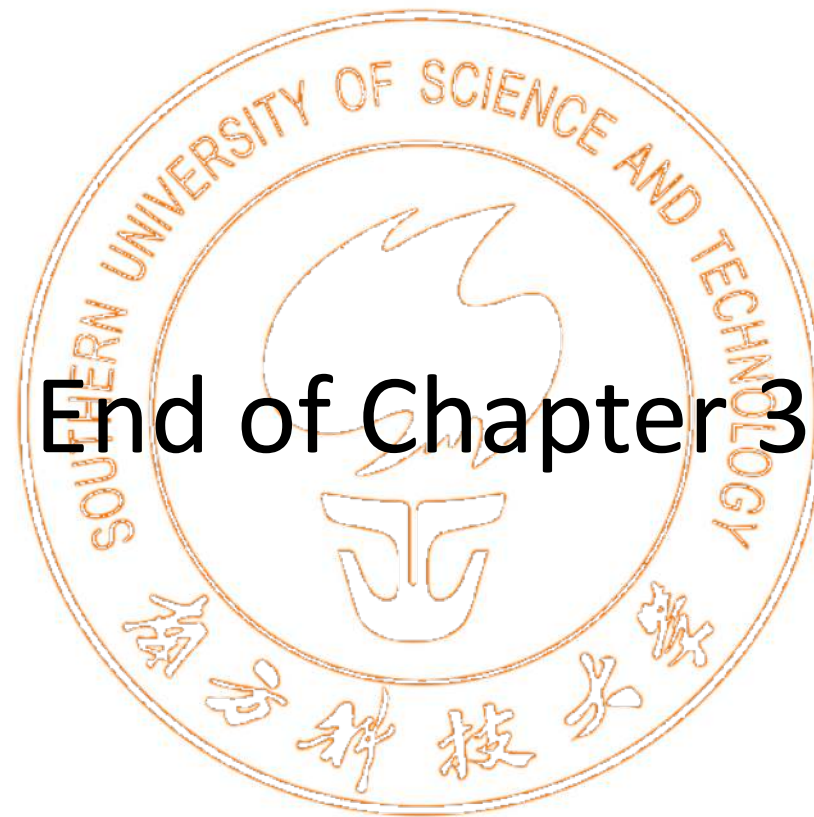


# Probability Theory and Statistics

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- Probability Theory move from known models to predictions about outcomes; Statistics move from outcomes to inference about models.
- Probability Theory provides the theoretical tools that Statistics builds on, and Statistics applies these tools to real-world data to draw conclusions and make predictions.
- Probability Theory can be thought as the “theory of the game” where you know all the rules.
- Statistics can be thought as the “detective work” where you figure out the rules by observing the game’s outcomes.





End of Chapter 3